



มหาวิทยาลัยขอนแก่น

วิทยา จริยา ปัญญา

KHON KAEN UNIVERSITY



HTML5 (Part 2)

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FACULTY OF ENGINEERING KHON KAEN UNIVERSITY

Agenda

- Add Figures
- Create general page structure
- Understand block-level and inline elements
- Add breaks and lines to your content



Figures

- HTML also offers a specific kind of image tag for figures.
- A figure is an image that could be moved around in your web page without dramatically altering the "flow" of the page



Millie Hughes-Fulford

```
1 <figure>
2   
3   <figcaption>Millie Hughes-Fulford</figcaption>
4 </figure>
```



Exercise1

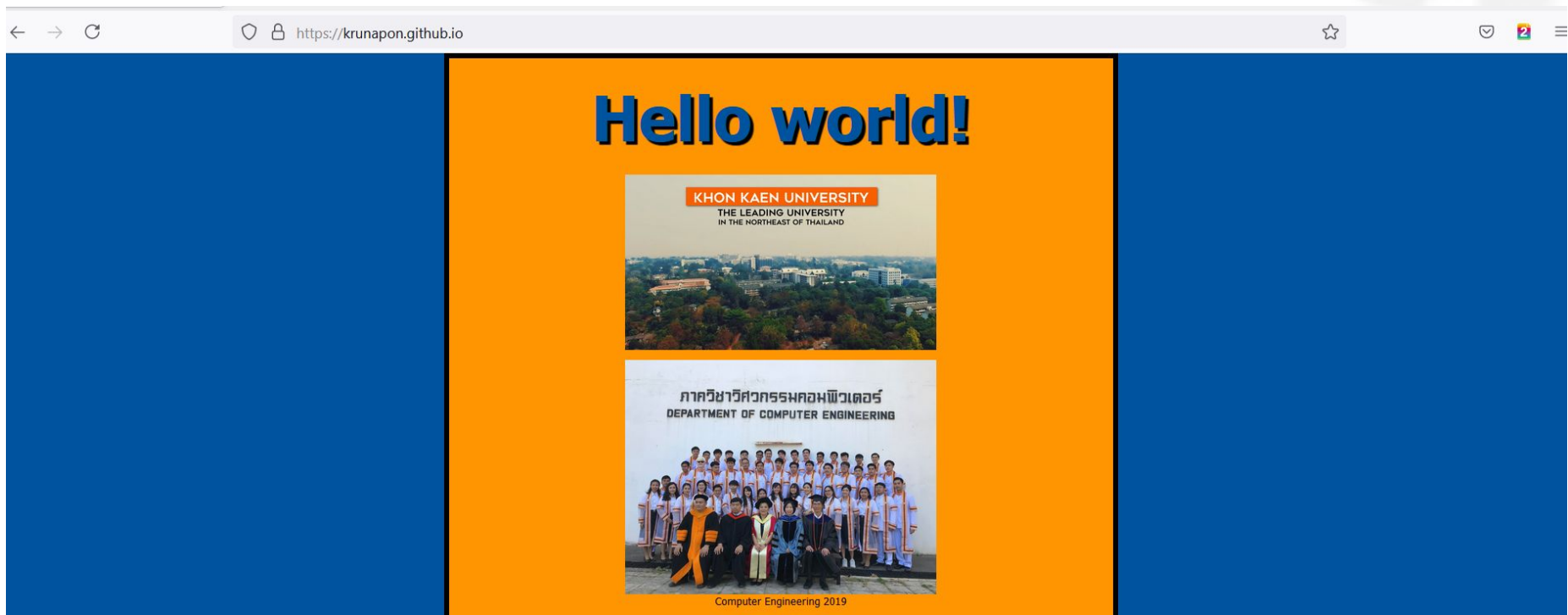
Using figure and figcaption and add an image to your <username>.github.io such that

- the new figure has some margin on top and bottom
- the new figure has caption at the centered below the image
- the new figure is centered at the page



Exercise1 sample output

<https://krunapon.github.io>



Web page structure

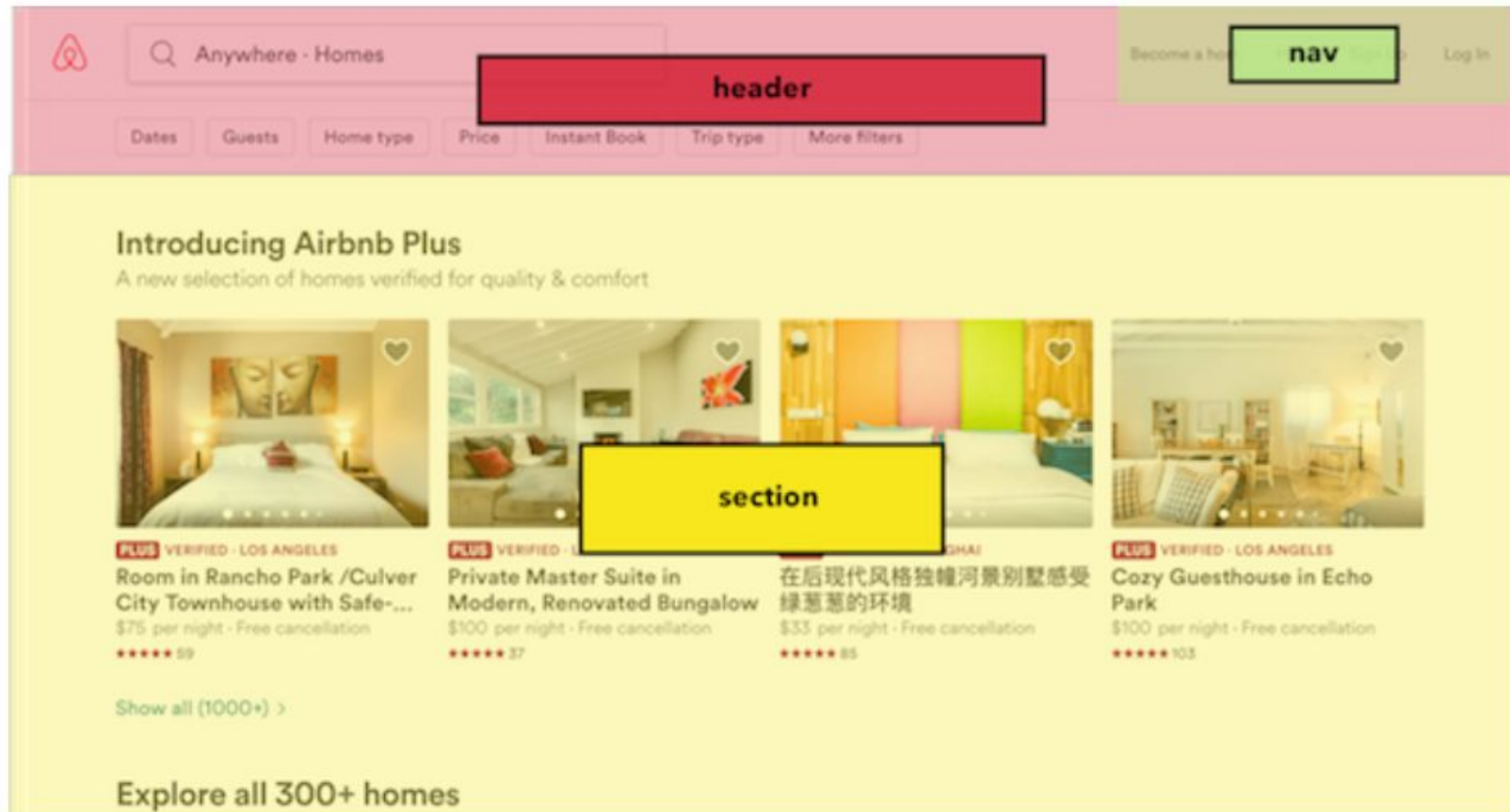
- When you build a house, you start by building a structure first, and then filling in the smaller elements, right?
 - The same is true when building a web page!
- What do the New York Times, Wikipedia, Airbnb, and Lady Gaga's website have in common?
 - They can all be broken down into predictable blocks



New York Times page structure



Airbnb page structure



Wikipedia page structure

The image shows a screenshot of the Wikipedia page for 'Duck'. The page is annotated with several colored boxes and labels to illustrate its structure:

- header**: A red box at the top center of the page content area.
- nav**: A green box on the left side of the page, containing navigation links like 'Main page', 'Contents', and 'Recent changes'.
- nav**: A green box at the top right of the page, containing links like 'Not logged in', 'account', and 'Log in'.
- article**: A purple box in the center of the page, covering the main text area.

The page content includes the Wikipedia logo, the title 'Duck', a summary paragraph, a detailed paragraph about ducks, and a sidebar with a list of contents and a scientific classification table.

Contents [hide]

- 1 Etymology
- 2 Morphology
- 3 Behaviour
 - 3.1 Feeding
 - 3.2 Breeding
 - 3.3 Communication
 - 3.4 Distribution and habitat
 - 3.5 Predators
- 4 Relationship with humans
 - 4.1 Domestication
 - 4.2 Hunting
 - 4.3 Cultural references
- 5 See also
- 6 References
- 7 External links

Scientific classification

Kingdom:	Animalia
Phylum:	Chordata
Class:	Aves
Order:	Anseriformes
Superfamily:	Anatoidea
Family:	Anatidae
Subfamilies	
see text	



Structural elements in HTML (1/2)

- `<header>`
 - The section at the top of a web page that often includes a logo and sometimes a nav
- `<nav>`
 - a set of menu items that allow a user to navigate to different pages on a site
- `<section>`
 - a general section of related content
- `<article>`
 - a piece of content that can be independently shared, like a blog post or newspaper article



Structural elements in HTML (2/2)

- `<footer>`
 - the section at the bottom of a page that often has additional links and perhaps social sharing icons
- `<aside>`
 - content that is complementary but not crucial to the page's content
- `<figure>`
 - a group image and caption that create an informative visual of some kind



Sample Usage of <header>, <nav>, <main>, <section>

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>My Travel Blog</title>
</head>
<body>

  <header>
    <h1>My Travel Blog</h1>
    <p>Exploring the world one
city at a time</p>
```

```
<nav>
  <a href="#bangkok">Bangkok</a>
  <a href="#tokyo">Tokyo</a>
  <a href="#paris">Paris</a>
</nav>
</header>
<main>
  <section id="bangkok">...</section>
  <section id="tokyo">...</section>
  <section id="paris">...</section>
</main>
</body>
</html>
```



Key Points of Using `<header>` and `<nav>`

- `<title>` is in `<head>` — sets the browser tab text
- `<header>` is in `<body>` — visible on the page, contains `<h1>` (the displayed page name)
- `<nav>` is nested inside `<header>` — common real-world pattern
- The `href="#bangkok"` links scroll to `<section id="bangkok">` further down
- `<main>` wraps **all primary content** — only one per page
- Each `<section>` has an `id` matching the `href` in `<nav>` — this is what makes anchor links work



Sample Usage of <figure>

```
<section id="bangkok">
<h2>Bangkok</h2>
<figure>

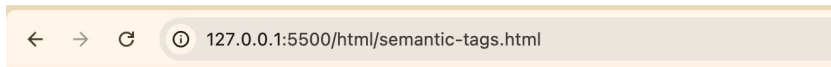
<figcaption>The Grand Palace,
Bangkok's most iconic landmark
</figcaption>
</figure>
<p>Bangkok is the capital city of
Thailand...</p>
</section>
```

Key points for <figure>:

- <figcaption> goes **inside** <figure>, either before or after
- alt and <figcaption> serve different purposes — alt is for screen readers/broken images, <figcaption> is visible text for all users
- One <figure> per image that needs a caption; decorative images don't need <figure>



Sample Output



My Travel Blog

Exploring the world one city at a time

[Bangkok](#) [Tokyo](#) [Paris](#)

Bangkok

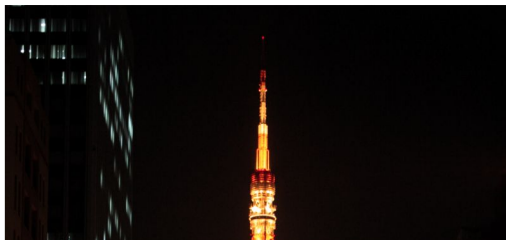


© WWF-Thailand/ Roengchai Kongmyang

The Grand Palace, Bangkok's most iconic landmark

Bangkok is the capital city of Thailand. It is the most populous city in the country.

Tokyo



Tokyo



Tokyo Tower, a famous landmark in Tokyo

Tokyo is the capital of Japan. It is the world's most populous metropolitan area.

Paris



Sample Usage of <aside>

<aside>

<h3>Travel Tips</h3>

<p>Best time to visit

Southeast Asia is

November to

February

when the weather is

cooler and drier.</p>

</aside>

Key points for <aside>:

- Placed **outside** <main> or inside it — both are valid
- Content is **related but not essential** to the main content
- Good candidates: tips, fun facts, quotes, related links



Sample Usage of <footer>

<footer>

<hr>

<p>Written by
Manee
Jaidee

</p>

<p>© 2026 My Travel
Blog</p>

</footer>

Key points for <footer>:

- Goes at the **bottom of <body>**, after <main> and <aside>
- © renders as © — a good HTML entity to introduce here
- Can also contain contact info, social links, or a back-to-top link



Sample Output

← → ↻ ⓘ 127.0.0.1:5500/html/semantic-tags.html

Paris



The Eiffel Tower, a famous landmark in Paris

Paris is the capital of France. It is known for its art, fashion, and culture.

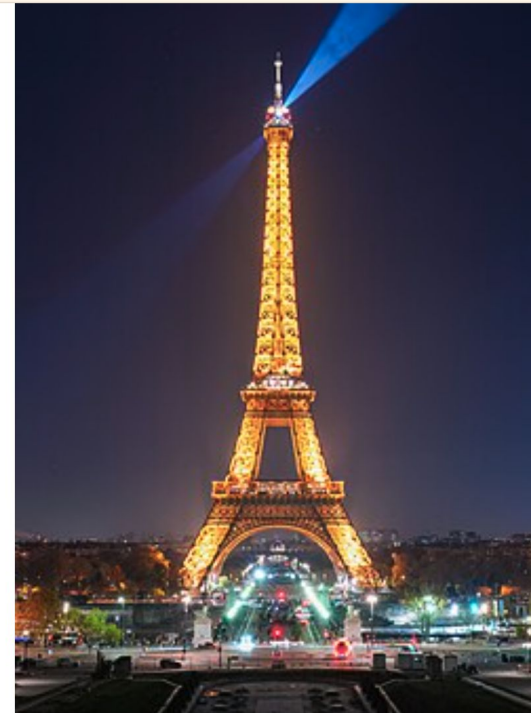
Travel Tips

Best time to visit Southeast Asia is **November to February** when the weather is cooler and drier.



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← → ↻ ⓘ 127.0.0.1:5500/html/semantic-tags.html



The Eiffel Tower, a famous landmark in Paris

Paris is the capital of France. It is known for its art, fashion, and culture.

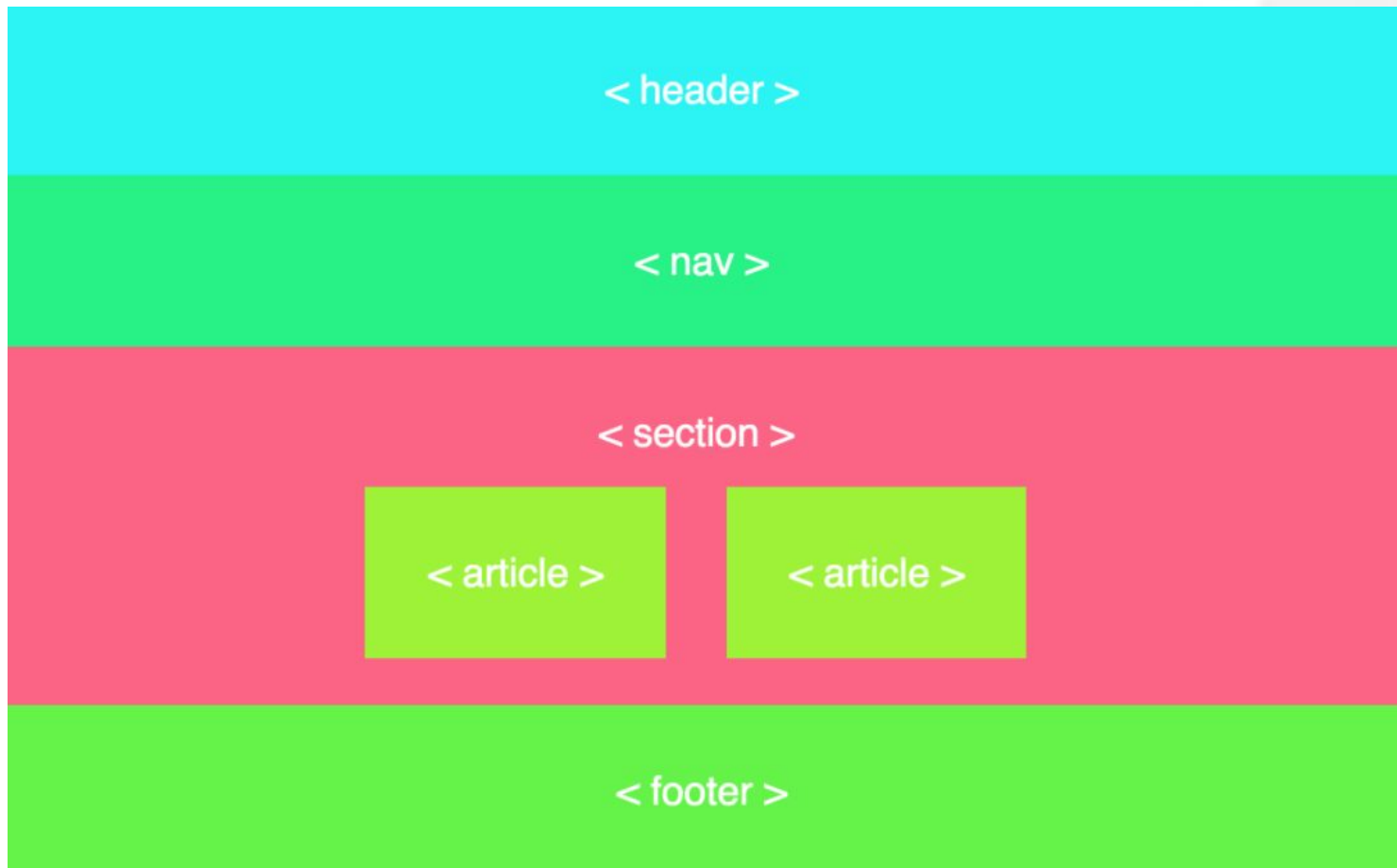
Travel Tips

Best time to visit Southeast Asia is **November to February** when the weather is cooler and drier.

Written by **Manee Jaidee**

© 2026 My Travel Blog

Possible desktop site structure



Possible mobile site structure



Agenda

- Add Figures
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- **Understand block-level and inline elements**
- Add breaks and lines to your content



Default behaviors of HTML elements (1/2)

- HTML elements have default behavior, via CSS behind the scenes, that controls their positioning
- Each paragraph expands to *take up the full width of the page on its own line* (greedy paragraphs)! However, the images all *appear in the same line*.

```
1 <p>Paragraph 1</p>
2 <p>Paragraph 2</p>
3 <p>Paragraph 3</p>
4 
5 
6 
```

Paragraph 1

Paragraph 2

Paragraph 3



Block-level elements vs. inline elements

- Every HTML element has a default block-level or inline behavior
- Paragraphs are block-level elements, which means that they block off a whole line for themselves
- Images are inline elements, which means they will automatically be placed next to one another on the same line



Block-level elements

Block-level elements that you've seen so far include:

- Headings
- Paragraphs (p)
- Lists and list items (ul, ol, li)
- Structuring elements (header, nav, section, article, aside, figure, footer)



Inline elements

Inline elements that you've seen so far include:

- Images (img)
- Emphasized text (em)
- Strong text (strong)
- Links (a)



Divs

- Divs are a block-level element used to group content together, not dissimilar to `<header>`, `<nav>`, `<section>`, `<article>`, `<footer>`, `<figure>`, and `<aside>` which you've already seen in this course
- Sometimes your content doesn't really fit any of these categories!
- In that case, you're encouraged to use divs to create block-level content groups.



Note: this is an editorial piece and does not reflect the official opinion of the Sun Journal.
For more information about our editorial team, click [here](#).

For more information about our editorial team, click [here](#).

Diam in arcu cursus euismod quis. Aliquet lectus proin nibh. Blandit massa enim nec dui nunc mattis enim ut tellus. Integer neque volutpat ac tincidunt vitae. Amet porttitor eget dolor morbi. Sed arcu risus. Orci phasellus egestas tellus rutrum tellus pellentesque tincidunt tortor.

```
1 <article>
2   <h1>Are self-driving cars worth the risk?</h1>
3   <div>
4     <p>Note: this is an editorial piece and does not appear in The New York Times Journal.</p>
5     <p>For more information about our editorial process, visit our website.</p>
6   </div>
7   <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Id cursus metus aliquam eleifend mi habitasse morbi leo a. Nisl vel pretium lectus quam id leo laoreet id commodo egestas diam.</p>
8   <p>Diam in arcu cursus euismod quis. Aliquet lectus accumsan sit amet nunc mattis enim ut tellus. Integer enim neque volutpat sed nulla morbi non arcu risus. Orci phasellus egestas tellus rutrum venenatis.</p>
9 </article>
```

- 
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Spans

- Spans are similar to divs, except they are inline elements. You might use them to group words together, since words appear inline and not each on their own line
- Imagine the newspaper above always prints its name in bolded light green
- You need to grab those two words — "Sun Journal" — to be able to apply CSS to them. Why not wrap them in a `<span>`?

```
1 <p>Note: this is an editorial piece and does not reflect the official opinion of the <span>Sun  
Journal.</span></p>
```

Note: this is an editorial piece and does not reflect the official opinion of the **Sun Journal**.

For more information about our editorial team, click [here](#).



The need for classes and ids (1/2)

- Let's say you're working on a news website, which is likely to contain paragraphs and paragraphs of text
- Some paragraphs will be within news articles, others might be disclaimer text at the bottom of the page, others will be short article blurbs
- Since they're all just `<p>` tags, you won't be able to apply CSS (appearance) changes to some paragraphs but not others.



The need for classes and ids (2/2)

- This is where classes and ids enter the scene.
- **Classes** and **ids** are custom attributes you can add to your elements in order to distinguish them from one another.
- **Classes apply to groups of elements, and ids apply to only one single element on an entire page**



The need for classes and ids (1/2)

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The need for classes and ids (2/2)

- This is where classes and ids enter the scene.
- **Classes** and **ids** are custom attributes you can add to your elements in order to distinguish them from one another.
- **Classes apply to groups of elements, and ids apply to only one single element on an entire page**



Classes

- Setting a class attribute is as simple as choosing the class name and adding the attribute to your element

```
1 <h1 class="breaking-news">Your heading here</h1>  
2 <p class="warning">Your paragraph text here</p>  
3 <p>More paragraph text is <span class="highlight">coming your way.</span></p>
```

In this very generic code example,

- The heading has a class of "breaking-news".
- The paragraph has a class of "warning".
- The span has a class of "highlight".



Sample usage of class

Lorem ipsum dolor sit amet, consectetur adipiscing elit. C blandit ut tellus.

Pellentesque ac enim vel neque volutpat luctus vitae qui pharetra odio eu, aliquet ipsum. Nullam luctus sem quis orci mi, tincidunt vitae eleifend in, fringilla ac tellus. Fusco commodo nec.

Watch out! Nunc vulputate sit amet enim ut efficitur. Sed

Maecenas quis nunc facilisis, cursus diam eget, bibendum vehicula. Vivamus urna turpis, fringilla eu efficitur ac, luctus

Integer vel quam consequat, accumsan nisi nec, consequat faucibus orci luctus et ultrices posuere cubilia Curae; Duis

Watch out! Mauris et quam nec felis mattis sagittis vel eu aliquam. Donec scelerisque non arcu id molestie. Nulla auctor in.

```
1 <p>Lorem ipsum dolor sit amet, consectetur  
laoreet et, blandit ut tellus.</p>  
2  
3 <p>Pellentesque ac enim vel neque volutpat  
pharetra odio eu, aliquet ipsum. Nullam lu  
mi, tincidunt vitae eleifend in, fringilla  
fringilla est commodo nec. </p>  
4  
5 <p class="warning">Watch out! Nunc vulputa  
viverra pretium.</p>  
6  
7 <p>Maecenas quis nunc facilisis, cursus di  
vehicula. Vivamus urna turpis, fringilla e  
8  
9 <p>Integer vel quam consequat, accumsan ni  
faucibus orci luctus et ultrices posuere c  
</p>  
10  
11 <p class="warning">Watch out! Mauris et qu  
tempor nibh in sollicitudin aliquam. Donec  
augue, non rhoncus dui auctor in.</p>
```



Ids

- Ids are just like classes — they are custom attributes that can be added to elements — except they'll only apply to **one element per page**
- Example

```
1 <p id="key-takeaway">Therefore, this is the one thing you should remember from the whole page.</p>
```



There will only be one key takeaway per page, so we're safe to use an id on this element instead of a class!



Naming ids and classes

- Name ids and classes based on context or function, not on appearance.
- For example, a class called "warning" could apply to scenarios where the content should have a warning function or context.
- A really bad class name for the same example would be "orange-background-paragraph" because it describes the content's pure appearance and not its function or context
 - Besides, what if you decide warning text should be yellow someday instead of orange?



The usage of ids and classes with CSS

HTML

```
consequat dolor. Vestibulum a  
faucibus orci luctus et ultr  
Curae; Duis ultrices nisi vel  
6 ▾ <p class="warning">Watch out!  
felis mattis sagittis vel ege  
nibh in sollicitudin aliquam.  
arcu id molestie. Nullam luct  
rhoncus dui auctor in.</p>  
7 ▾ <p>Nam sodales efficitur dolo  
a lacus eu nibh tempor fermer  
8 ▾ <p id="key-takeaway">Therefore  
you should remember from the
```

CSS

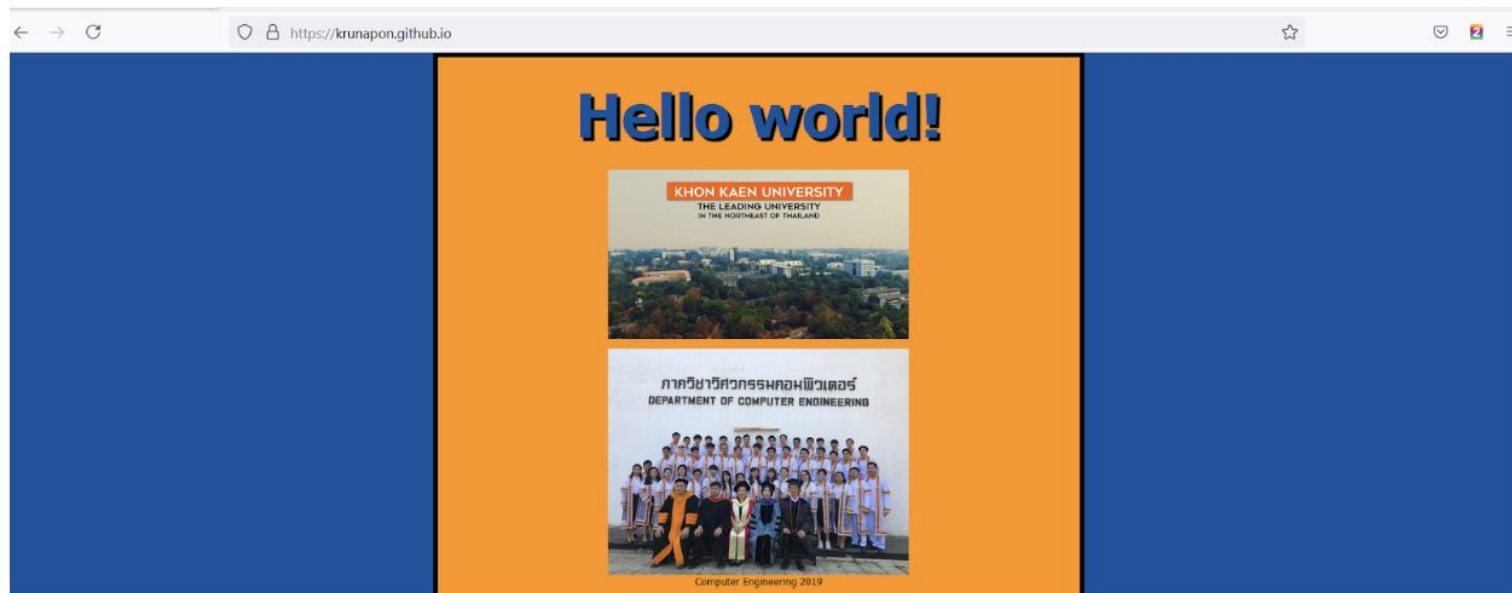
```
7  
8 ▾ .warning {  
9     background-color: #FF7E6B;  
10    padding: 10px;  
11    border-radius: 4px;  
12 }  
13  
14 ▾ #key-takeaway {  
15     background-color: #ffffff;  
16     font-size: 18px;  
17     padding: 10px;  
18     border: 4px dashed black;  
19     border-radius: 4px;  
20 }
```



Exercise 2 (Part 1)

Using figure and figcaption and add an image to your <username>.github.io such that

- the new figure has some margin on top and bottom
- the new figure has caption at the centered below the image
- the new figure is centered at the page



Exercise2 (Part 2)

- Modify the page <your_username>.github.io
- Add the **id 'heading'** to the first paragraph and apply CSS style to the element with the id such that **the element has the text color blue**
- Add the **class 'popular'** to the first item and the third item of the list and apply CSS style to the items by adding a red dotted border with **line=4px** around the items



Exercise2 (Part 2) output

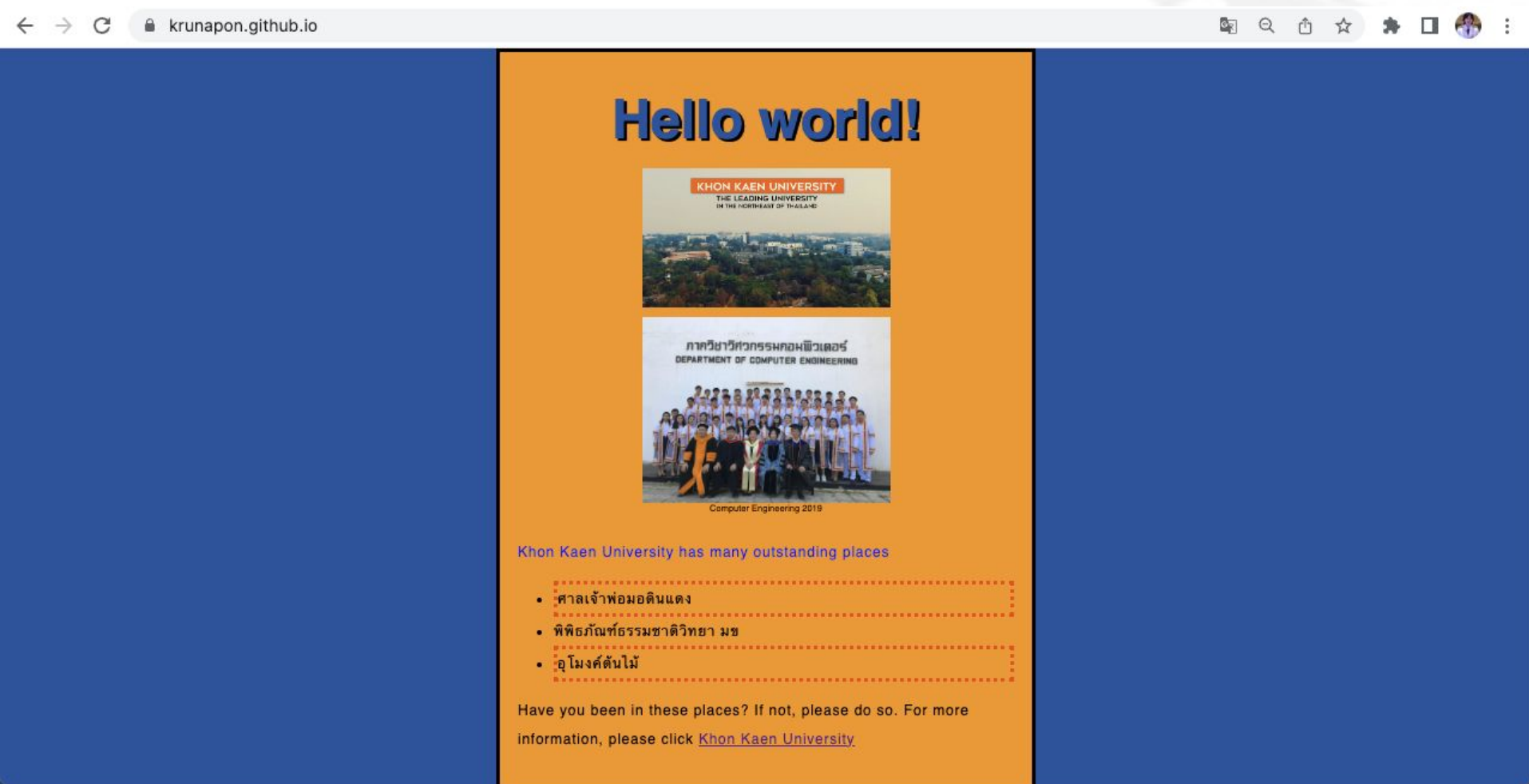
Khon Kaen University has many outstanding places

- ศาลเจ้าพ่อมอดินแดง
- พิพิธภัณฑ์ธรรมชาติวิทยา มข
- อุโมงค์ต้นไม้

Have you been in these places? If not, please do so. For more information, please click [Khon Kaen University](#).



Exercise 2 output



Agenda

- Add Figures
- Create general page structure
- Understand block-level and inline elements
- **Add breaks and lines to your content**



Using line breaks or horizontal rules

- There are easy ways to break up the structure of your page in order to separate themes or ideas without creating classes or ids.
- You can use line breaks or horizontal rules (i.e., lines) to do

New York's best coffeeshops

New York has some great coffeeshops. Make sure not to miss the hot new addresses opening up left and right!

Manhattan

The Bean

824 Broadway
New York, NY 10003

There are multiple Bean locations in the city. It's a good place to get work done because they're open late.

Third Rail

240 Sullivan St
New York, NY 10012

Coffee and donuts! What more could you want?

Black Fox Coffee Co.

70 Pine St
New York, NY 10270

Black Fox has an incredible bean selection and nice snacks.

Stumptown

18 W 29th St
New York, NY 10001

Does this even need explaining?



Example of the use of line breaks

- To create line breaks in HTML, use the `<br/>` (`<br></br>`) tag.

```
1 <h3>Toby's Estate</h3>
2 <p class="address">
3     125 N 6th St
4     <br>
5     Brooklyn, NY 11249
6 </p>
```

Toby's Estate
125 N 6th St
Brooklyn, NY 11249

- If we use two `<p>` elements

125 N 6th St
Brooklyn, NY 11249



Horizontal rules

- To create a line, or a "horizontal rule" in HTML, simply use an `<hr>` tag

```
1 <h3>Manhattan</h3>
2 ...
3 <hr>
4 ...
5 <h3>Brooklyn</h3>
```

Manhattan

[coffeeshops here]

Brooklyn

[coffeeshops here]



References

- <https://openclassrooms.com/en/courses/5265446-build-your-first-web-pages-with-html-and-css>

